

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — EXAMINATION QUESTION PAPER

Subject: Data Structure

Time: 1.5 Hours

Semester / Programme: 3rd Semester

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

1(a) How many different Binary Search Trees (BST) can be constructed with 4 distinct keys? [1]

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

1(b) Write the postfix expression for the following infix expression: $A + B \times C - D / E$ [1]

1(c) Define a sparse matrix and state its primary storage advantage in computer memory. [1]

1(d) State the worst-case time complexity of searching an element in a Binary Search Tree (BST) of height h . [1]

$$T(n) = O(h)$$

1(e) What is the maximum number of nodes in a binary tree of height h (assuming root is at height 0)? [1]

$$N_{max} = 2^{h+1} - 1$$

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

Q.2. (a) Write a C/C++ function to implement PUSH and POP operations on a Stack represented using a Singly Linked List. [3]

(b) Explain dynamic storage management and how garbage collection helps in memory reclamation. [2]

Q.3. (a) Construct a binary tree given the following traversal sequences:
Preorder: A B D E C F
Inorder: D B E A F C [3]

(b) Write a recursive function to find the height of a given binary tree. [2]

$$\text{Height}(T) = 1 + \max(\text{Height}(L), \text{Height}(R))$$

- Q.4. (a)** Perform operations on polynomials: Represent polynomial $P_1 = 4x^3 + 3x^2 + 5$ using a singly linked list and write an algorithm to add two polynomials. [3]
- (b)** Explain the advantages of Circular Linked Lists over Singly Linked Lists with a suitable example. [2]
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- Q.5. (a)** Write a non-recursive pseudo-code/algorithm to traverse a Binary Tree in In-order using an explicit stack. [3]
- (b)** Convert the following infix expression into its prefix equivalent: $(A + B) \times C - (D / E)$ [2]
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